CSCI 8 (Data Science) **Lecture 18**

Decisions & Uncertainty

Recap from Lectures 16 and 17

- **One Binary Category**

Ex1: Swain vs. Alabama
* _____ category: Black people on the jury panel (vs. not Black)
* Alternative hypothesis: _____ (the proportion is _____ expected)
* Test statistic: Empirical percentage (sample proportion) of Black jurors

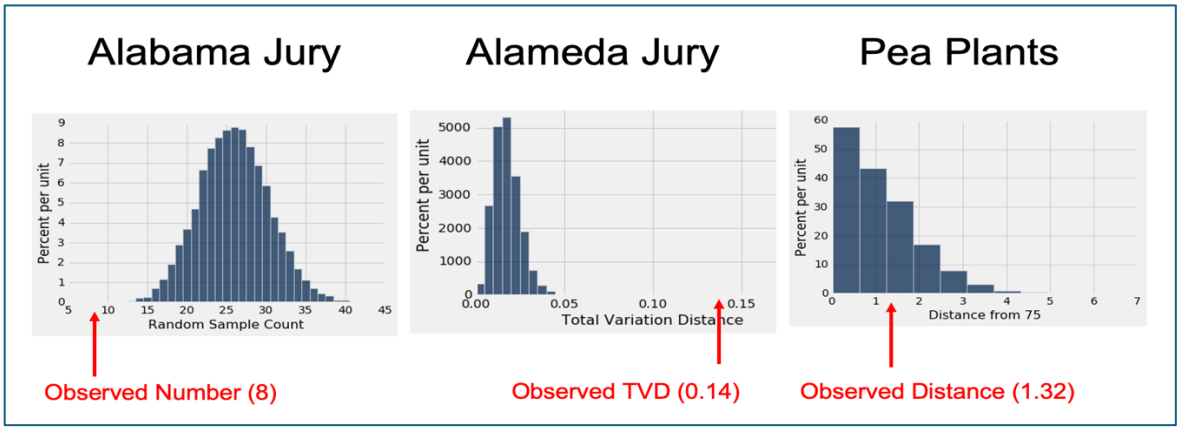
Ex2: Mendel and Pea Flowers
* _____ category: Purple flowers (vs. not purple)
* Alternative hypothesis: _____ (the proportion is _____ from expected)
* Test statistic: |empirical percentage – null percentage|

- **Multiple Categories**

Ex3: Jury Selection in Alameda County
* _____ categories: Asian, Black, Latino, White, and Other
* Alternative hypothesis: Two-sided (the proportions are *different* from expected)
* Test statistic: _____

TVD = $\frac{1}{2}\sum$ |empirical percentage – null percentage|

Our Decisions



Alabama Jury → Data is _____ with the model.
Alameda Jury → Data is _____ with the model.
Pea Plants → Data is _____ with the model.

What if visualizations are not obvious enough to make a decision? → Use the _____, which is what we are learning today.

Example 4: The GSI's Defense

The Problem

- A large statistics class is divided into 12 discussion sections.
- Graduate Student Instructors (GSIs) lead the sections.
- After the midterm, students in Section 3 noticed that their average score in their section is lower than the other sections.

Setting Up Hypotheses

Null Hypothesis (GSI's position):

Section 3's average score is **not unusually low.** In other words, if we had randomly selected a section from the class, we could have gotten an average like Section 3's just by chance.

Alternative Hypothesis:

Section 3's average is **too low to be explained by randomness alone**. Something beyond random variation may be affecting the scores.

Choosing a Test Statistic

Test Statistic: _____

→ Next: Open "Demonstration Notebook – Decisions & Uncertainty" (lec18) → go to "The GSI's Defense".

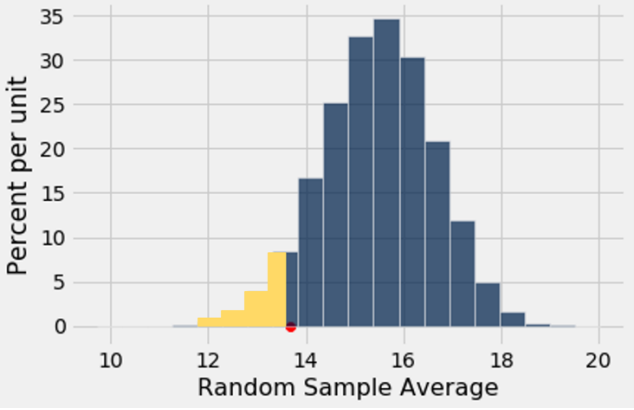
Two Approaches for Making Decisions in Hypothesis Testing

1. The p-value approach:

A **p-value** is:

- the probability of
- observing a test statistic
- as extreme as or more extreme than the one observed,
- assuming the null hypothesis is true;
- equivalently, it is the area in the tail of the null distribution in the direction that supports the alternative hypothesis.

Empirical distribution of the test statistic under the null hypothesis



For a left-tailed test, p-value
≈ (number of simulated test statistics ≤ observed test statistic) ÷ (total number of simulated test statistics)

For a right-tailed test, p-value
≈ (number of simulated test statistics ≥ observed test statistic) ÷ (total number of simulated test statistics)

Conventions About Inconsistency

- **"Inconsistent with the null":** The test statistic is **<u>in the tail</u>** of the empirical distribution under the null hypothesis

- **"In the tail," first convention:**
  - If the area in the tail is less than 5%,
  - the result is "*statistically significant*"

- **"In the tail," second convention:**
  - If the area in the tail is less than 1%,
  - the result is "**highly** *statistically significant*."

Note: "Statistically significant" means that the result you observed in your data is unlikely to have happened just by random chance, assuming the null hypothesis is true.

→ Next: Open "Demonstration Notebook – Decisions & Uncertainty" (lec18) → go to "Our Decision". (to perform the p-value approach and assess statistical significance)

<u>Making a Decision</u>

p-value for this test ≈ _____

significance level (threshold) = _____

Decision: Since the p-value is _____ than 5%, the observed test statistic _____ statistically significant**.** We conclude that the observed value is _____ with the null hypothesis, which states that the midterm average score of Section 3 _____ unusually low.

2. The critical value approach:

A **critical value** is the borderline (cutoff) value that separates the threshold area (such as 5% or 1%) from the rest of the sampling distribution.

Decision criterion: If the observed test statistic is more extreme than the critical value in the direction specified by the alternative hypothesis, the data provide evidence in favor of the alternative hypothesis.

→ Next: Open "Demonstration Notebook – Decisions & Uncertainty" (lec18) → go to "Approach 2 (Compare observed test statistic to critical value)".
(to perform the critical value approach)

Recap: The basic structure of testing a hypothesis

1. Setting up your question:
    - _____ Hypothesis: For a fully specified chance model under which you can simulate the data, the hypothesis is that the data were generated as if by random selection, according to the particulars of the chance model.
    - _____ Hypothesis: states that there is some process other than the chance model that generated the data.
2. Choose a _____:
    - A value that can be computed for the observed data and also for random samples from the chance model.
    - When this value is computed for the observed data, it is called the _____.
3. Simulate under the _____:
    - Use random sampling to create an _____ distribution of the test statistic, assuming that the chance model is true. This empirical distribution approximates the sampling distribution of the test statistic under the null hypothesis.
4. Conclusion:
    - **Using P-value Approach**: Compute the p-value, which is the probability (area) beyond the _____ in the direction specified by the alternative hypothesis. Compare the p-value to the significance level (5% or 1%). If the p-value _____ significance level, the data are consistent with the _____.

    - **Using Critical Value Approach**: Determine the critical value (the 5% or 1% cutoff point) from the sampling distribution. Compare it to the observed test statistic. If the observed test statistic is _____ than the critical value in the direction specified by the alternative hypothesis, the data are consistent with the alternative hypothesis.